

Question				Marking details	Marks available					
					AO1	AO2	AO3	Total	Maths	Prac
	(b)			Substitution: $P = VI = 9 \times 4.5$ or $P = I^2 R = 4.5^2 \times 2$ ecf from (a)(ii) (1) $= 40.5$ [W] (1) Either for the 3rd mark: In 10 s, $E = P \times t = 40.5 \times 10 = 405$ [J] so disagree Or: $P = \frac{E}{t} = \frac{45}{10} = 4.5$ [W] so disagree Alternative: $P = \frac{45}{10}$ (1) $= 4.5$ [W] (1) Either for the 3rd mark: $I = \frac{P}{V} = \frac{4.5}{9} = 0.5$ [A] so disagree Or: $V = \frac{P}{I} = \frac{4.5}{4.5} = 1$ [V] so disagree Conclusion must be present to award 3 marks			3	3	3	3
				Question 7 total	3	6	3	12	11	12